

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Minutes

QUESTIONS: 30

Time Duration: 30

Total Marks:30

Annexure A

MCQ

Code of Conduct:

I __B.SREENITHA-192312203____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B.Sreenitha

Signature of the student:

Annexure A

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
- b) $\neg P \vee Q$
- c) $\neg Q \rightarrow \neg P$ only
- d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses)
- c) A conjunction of literals only
- d) A sequence of implications

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
- b) Every entity that is a student passes
- c) If something passes, it is a student
- d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

6. What is the core function of a logical agent in AI?

- a) Reacting only to current percepts
- b) Using logic and knowledge representation to reason and make decisions
- c) Performing physical actions without internal representation
- d) Relying solely on sensors for navigation

7. Which component of a logical agent is responsible for storing facts and rules about the world?

- a) Inference Engine
- b) Perception Unit
- c) Knowledge Base (KB)
- d) Action Selector

8. What does a knowledge-based agent do immediately after obtaining a percept from the environment?

- a) Selects an action
- b) Executes the action
- c) Updates the Knowledge Base with the new percept

d) Reasons to derive new information

9. Which operation is used to add new information to a Knowledge Base?

a) ASK b) TELL c) PERCEIVE d) UPDATE

10. What is a proposition in formal logic?

- a) A command or request
- b) A declarative statement that is either true or false
- c) A question regarding the environment
- d) A letter representing a variable

11. Which of the following is NOT a proposition?

- a) "The sun rises in the east."
- b) " $5 + 3 = 10$."
- c) "Close the door."
- d) "Chennai is in India."

12. Which logical operator is represented by the symbol $\neg$?

- a) AND
- b) OR
- c) NOT
- d) Implication

13. The AND ($\wedge$) operator is true only when:

- a) At least one proposition is true
- b) Both propositions are true
- c) Both propositions have different truth values
- d) Neither proposition is true

14. A proposition that is always true, regardless of the truth values of its variables, is called a:

- a) Contradiction b) Tautology c) Contingency d) Equivalence

15. In an implication ($P \rightarrow Q$), the statement is false only when:

- a) P is false and Q is true
- b) P is true and Q is false
- c) Both P and Q are false
- d) Both P and Q are true

16. What is the number of rows in a truth table for a logical expression with 3 propositions?

- a) 4 b) 6 c) 8 d) 9

17. Which of these represents one of De Morgan's Laws?

- a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- b) $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$

- c) $\neg(\neg P) \equiv P$
- d) $P \vee \neg P \equiv \text{True}$

18. A proposition that is sometimes true and sometimes false is known as a:

- a) Tautology b) Contradiction c) Contingency d) Implication

19. Which logical connective is true if and only if both propositions have the same truth value?

- a) Implication ($\rightarrow$) b) Biconditional ($\leftrightarrow$) c) OR ($\vee$) d) NOT ($\neg$)

20. According to operator precedence, which operator should be evaluated first?

- a) AND ($\wedge$) b) OR ($\vee$) c) NOT ($\neg$) d) Implication ($\rightarrow$)

21. What is the process of deriving a conclusion from given premises called?

- a) Perception b) Representation c) Inference d) Predication

22. Which inference rule follows the form: If P, then Q; P; Therefore, Q?

- a) Modus Tollens b) Modus Ponens c) Hypothetical Syllogism d) Resolution

23. Which valid reasoning pattern is represented by: $P \rightarrow Q$; $\neg Q$; Therefore, $\neg P$?

- a) Modus Ponens b) Disjunctive Syllogism c) Modus Tollens d) Addition

24. What can be concluded from the premises $(P \rightarrow Q)$ and $(Q \rightarrow R)$ using Hypothetical Syllogism?

- a) $P \wedge R$ b) $P \vee R$ c) $P \rightarrow R$ d) $\neg P \rightarrow \neg R$

25. The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

- a) Simplification b) Conjunction c) Disjunctive Syllogism d) Resolution

26. Which rule allows you to conclude "P" from the premise "P and Q"?

- a) Addition b) Simplification c) Conjunction d) Modus Ponens

27. The inference rule Resolution follows which symbolic form?

- a) $P \therefore P \vee Q$ b) $P, Q \therefore P \wedge Q$ c) $P \vee Q, \neg P \vee R \therefore Q \vee R$ d) $P \rightarrow Q, Q \rightarrow R \therefore P \rightarrow R$

28. Which of the following is an example of the "Addition" rule?

- a) P and Q, therefore P
- b) P, therefore P or Q
- c) P, Q, therefore P and Q
- d) P or Q, not P, therefore Q

29. "If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

- a) Denying the Antecedent b) Affirming the Consequent c) Modus Ponens d) Double Negation

30. What is the name of the invalid pattern: $P \rightarrow Q$; $\neg P$; Therefore, $\neg Q$?

a) Affirming the Consequent b) Denying the Antecedent c) Modus Tollens d) Hypothetical Syllogism

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

30

30

8

- ① b) ☒ knowledge base + logical inference.
- ② d) ☒ Both b and c.
- ③ b) ☒ conjunction of disjunctions.
- ④ b) ☒ Every student passes.
- ⑤ b) ☒ make two expressions syntactically identical.
- ⑥ b) ☒ using logic and knowledge representation
- ⑦ c) ☒ knowledge Base (KB)
- ⑧ c) ☒ updates the knowledge Base.
- ⑨ b) ☒ TELL
- ⑩ b) ☒ Declarative statement that is true or false.
- ⑪ c) ☒ "close the door"
- ⑫ c) ☒ NOT
- ⑬ b) ☒ Both Propositions are true.
- ⑭ b) ☒ Tautology
- ⑮ b) ☒ P is true and Q is false.

- | | | |
|-----|------|---|
| 16) | c) ✓ | 8 |
| 17) | a) ✓ | $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ |
| 18) | c) ✓ | contingency |
| 19) | b) ✓ | Biconditional ($\leftrightarrow$) |
| 20) | c) ✓ | NOT ($\neg$) |
| 21) | c) ✓ | Inference |
| 22) | b) ✓ | Modus Ponens |
| 23) | c) ✓ | Modus Tollens |
| 24) | c) ✓ | $P \rightarrow R$ |
| 25) | c) ✓ | Disjunctive Syllogism |
| 26) | b) ✓ | Simplification |
| 27) | c) ✓ | $P \vee Q, \neg P \vee R \therefore Q \vee R$ |
| 28) | b) ✓ | P , therefore $P \vee Q$ |
| 29) | b) ✓ | Affirming consequent |
| 30) | b) ✓ | Denying the Antecedent. |